A10: Independent Adversarial Audit
Positive Saddle-Node and Stable Coexistence

Directed Cyclic Networks research project

3 October 2026 Exact audited version: P10 v1.0

FOLD GEOMETRY / SPECTRUM VERDICT:	**VERIFIED**
FULL-SYSTEM SADDLE-NODE VERDICT:	**VERIFIED**
BRANCH STABILITY / BISTABILITY VERDICT:	**VERIFIED**
COEXISTENCE SET / COUNTEREXAMPLE VERDICT:	**VERIFIED**
BASIN / MANIFOLD VERDICT:	**VERIFIED**
SCOPE / DEPENDENCY VERDICT:	**VERIFIED**
OVERALL VERDICT:	**VERIFIED**

Required corrections: none. No mathematical or scope-changing wording issue was found in the exact authenticated source. All 707 source lines are covered by the range ledger below. This verdict rests on independent analytical reconstruction, including the real transverse implicit-function argument and the full local two-basin cap-exit proof. Neither the supplied validation package nor this audit's finite computations is used as proof.

Strongest verified conclusion. For every canonical $a > k+h$ and every admissible (n, q), the positive fold is a nondegenerate full-system saddle-node with a simple uniform zero and strictly stable transverse spectrum. Sufficiently below the fold, the origin and upper positive equilibrium are local exponential sinks, while the lower equilibrium is an index-one saddle. This configuration persists openly in all five physical parameters. Near that saddle, its stable disk is exactly the common boundary, relative to a sufficiently small positive neighborhood, of the two sink basins. The global synchronous-ray threshold also holds throughout the two-positive-root interval. No full global basin classification follows.

This audit does not edit P10, its draft archive, or any project lifecycle record. P10 remains DRAFT pending administrative closure in its original development chat. No P11 proof or literature survey is undertaken.

1 Current-file authentication and dependency contract

The manifest's final CURRENT P10 entry selects `proofs/P10_saddle_node_and_bistability.tex`, version P10 v1.0, with the full-file hash below. The exact file has 32,560 bytes and 707 lines. The audit uses the complete draft archive and matching working project at `output/p10/cyclic_network_project/`, not an earlier placeholder.

Audited full-file SHA-256:
`a28ea2b4ca96f381f06fa82849d4289fcc0e9c41e646ca66595621922f385a07`

Mathematical body SHA-256, line 19 through EOF:
`5cfd8afb91b7f23575e1b4bd9c7c28f3b433affefae9bb0d5f7700d030fcba81`

Original P10 draft ZIP SHA-256:
`a3b8ba33e16095a8880637bdfe74a2fcb2e26bcc1ada8d941efe85cbccf5c42e`

Authenticated P09-audited baseline ZIP SHA-256:
`db6344c5ed7c96f244f99eefa589ae8631c3aaf15278577e607e246026b22bf2`

Both existing P10 draft ZIP copies are byte-identical. ZIP CRC checks pass; all 176 files have the sole root `cyclic_network_project/` and match the working copy byte-for-byte. Relative to the authenticated P09-audited archive, exactly five members differ: P10, manifest, registry, compendium source and compendium PDF. No baseline member is removed or added.

All ten requested control files and all P01–P05, P07–P09 and matching audit sources A01–A05, A07–A09 are present. They are inventoried with complete hashes, sizes and line counts in `A10_support/input_inventory.json`. The open-decisions record was also authenticated for the B03 scope history. All eight prerequisite mathematical-body hashes match their current headers, and every prerequisite proof/audit is byte-identical to the P09-audited baseline. A01/A02 E01 and A05/A08 E01 are already closed. A03/A04/A07/A09 require no correction. A02's historical missing-source header is superseded by its recorded source-restoration closure; the actual source is present. A04/A05 historical numerical-provenance limitations remain unchanged and nonblocking. Historical DRAFT/TODO text does not supersede the current manifest and closure records.

1.1 Actual inputs, rather than header declarations alone

The dependency map retains exactly

$$\mathrm{P02}, \mathrm{P03}, \mathrm{P04}, \mathrm{P09} \longrightarrow \mathrm{P10}, \qquad \mathrm{P01\ inherited.}$$

P02 supplies invariant synchrony, the scalar equation, root labels and algebraic fold locus. P03 supplies the original Jacobian and origin spectrum. P04 supplies Fourier phases, complete labeled spectrum, multiplicities, normality and linearized stability implications. P09 supplies identification of the origin branch and the cubic-degenerate limit. P01 supplies the smooth flow, uniqueness, orthant invariance and global nonnegative existence and boundedness. The real IFT, hyperbolic persistence, smooth stable/unstable manifold theorem and ordinary linearized stability are standard local tools. The invariant center manifold is exhibited explicitly; it is not obtained from a forbidden project module.

P05/P07/P08 are authenticated context only. Every P10 assertion was checked without their selection/comparison results. References to P11 are future scope pointers, not mathematical premises or additions to the frozen DAG. The frozen S4 supporting role permits stable coexistence and excludes a complete global basin claim; P10's explicit local theorem respects that boundary. Claim rows C070/C071 and P10-D1–D10 are DRAFT, with C147 refuted only in its universal origin/upper-coexistence interpretation. No novelty verdict is changed.

2 Layer A: fold geometry and complete spectrum

2.1 Original restriction, notation and exact roots (checks 3–5)

From the canonical field

$$F_i(x) = ax_i - kx_i^2 + \frac{cx_{i-1}}{1+x_i} - \frac{gx_i}{1+x_{i+q}} - hx_ix_{i-q}, \tag{1}$$

substitute $x = r\mathbf{1}$, $r > -1$. All five terms must be retained even when neighbor offsets coincide. Each component equals

$$f(r,g) = ar - (k+h)r^2 + \frac{(c-g)r}{1+r} = \frac{r}{1+r}\{\mu + br - Ar^2\}, \quad A = k+h, \quad b = a-A, \quad \mu = a+c-g.$$

Thus synchrony is invariant by uniqueness. With $b > 0$,

$$g_f = a + c + \frac{b^2}{4A} = c + \frac{(a+A)^2}{4A}, \qquad r_f = \frac{b}{2A}, \qquad s_f = 1 + r_f = \frac{a+A}{2A}.$$

The second expression for g_f follows from $4Aa + (a-A)^2 = (a+A)^2$; also $g_f - g_c = b^2/(4A) > 0$. P02's amplitudes $s_\pm$ are P10's $r_\pm$. P02's fold amplitude s_f is P10's r_f, whereas P10's s_f is the denominator. The source reconciles these explicitly in lines 31–36; no formula imports the wrong meaning.

Let $u = r - r_f$ and $\nu = g_f - g$. Since $\mu = \nu - b^2/(4A)$, $br_f - Ar_f^2 = b^2/(4A)$ and $b - 2Ar_f = 0$,

$$\mu + b(r_f + u) - A(r_f + u)^2 = \nu - Au^2, \tag{2}$$

$$\dot{u} = \frac{r_f + u}{s_f + u}(\nu - Au^2). \tag{3}$$

These are identities, not approximations. On $|u| < r_f/2$, neither prefactor vanishes. For $\nu < 0$ the polynomial is negative for every real u, so there is no nonzero synchronous equilibrium, including near the fold. At $\nu = 0$ there is one repeated positive root. For $\nu > 0$ the two distinct roots are $r_\pm = r_f \pm \sqrt{\nu/A}$; they are both positive exactly when $0 < \nu < Ar_f^2 = b^2/(4A)$. A sufficiently small positive ν places both inside the chosen state neighborhood. The ever-present origin is excluded by that neighborhood; it is not removed from the full system.

2.2 Physical fold identity and phase signs (checks 6–7)

Differentiating (1) coordinate by coordinate at synchrony gives

$$d = a - 2kr - \frac{cr}{(1+r)^2} - \frac{g}{1+r} - hr, \quad C = \frac{c}{1+r}, \quad G = \frac{gr}{(1+r)^2}, \quad H = hr.$$

For $(Pz)_j = z_{j-1}$, the $i-1$ input is CP, the $i+q$ loss derivative is $+GP^{-q}$, and the $i-q$ bilinear derivative is $-HP^q$. Thus

$$J = dI + CP + GP^{-q} - HP^q, \qquad \lambda_m = d + Ce^{-\mathrm{i}\theta_m} + Ge^{\mathrm{i}q\theta_m} - He^{-\mathrm{i}q\theta_m}, \quad \theta_m = 2\pi m/n.$$

The sign on G is positive because differentiating the reciprocal loss denominator reverses its sign. The imaginary H contribution is positive because $-H$ multiplies a negative sine. Offset collisions add coefficients.

The decisive fold identity follows directly from physical values:

$$g_f = c + As_f^2, \qquad \frac{g_f}{s_f^2} - h = A - h + \frac{c}{s_f^2} = k + \frac{c}{s_f^2} > 0.$$

Multiplying by $r_f > 0$ proves

$$\boxed{G_f - H_f = r_f\left(k + \frac{c}{s_f^2}\right) = r_f\left(k + \frac{4A^2c}{(a+A)^2}\right) > 0.}$$

At a nonzero equilibrium $f_r = r(b - 2Ar)/(1+r)$; hence $\lambda_0^f = f_r(r_f, g_f) = 0$ and $d_f = -C_f - G_f + H_f$. Subtracting this row sum gives

$$\operatorname{Re}\lambda_m^f = C_f(\cos\theta_m - 1) + (G_f - H_f)(\cos(q\theta_m) - 1),$$
$$\operatorname{Im}\lambda_m^f = -C_f\sin\theta_m + (G_f + H_f)\sin(q\theta_m).$$

2.3 Automatic gap and simple zero (checks 8–9)

For $1 \le m < n$, $j = \min(m, n-m)$ satisfies $2\pi j/n \in [2\pi/n, \pi]$. Cosine is decreasing there, so

$$\cos\theta_m \le \cos(2\pi/n) < 1, \qquad \cos(q\theta_m) \le 1.$$

Consequently

$$\operatorname{Re}\lambda_m^f \le -\delta_f < 0, \qquad \delta_f = \frac{c}{s_f}\{1 - \cos(2\pi/n)\}.$$

For odd n the last angle is below π; for even n it may equal π. Both cases obey the same inequality. At $q = 1$ or $q = n-1$, combined shifts still have the displayed spectrum. At even n, $q = n/2$, the two q shifts coincide but their difference is retained. If $\gcd(n, q) > 1$, some nonuniform labels can have $\cos(q\theta_m) = 1$; the strictly negative first term still provides the entire margin. No coprimality assumption is needed.

The unitary Fourier matrix has n independent columns and diagonalizes J_f. Exactly one diagonal entry is zero and all others have negative real part. Thus zero is the only imaginary-axis eigenvalue, has algebraic and geometric multiplicity one, and has real eigenspace $\operatorname{span}\{\mathbf{1}\}$. Coincidences among stable numerical eigenvalues do not change this conclusion. The gap is positive for each fixed n; it need not remain bounded below as $n \to \infty$. **FOLD GEOMETRY / SPECTRUM VERDICT: VERIFIED.**

3 Layer B: full-system saddle-node

3.1 Real transverse IFT and equilibrium completeness (checks 10–11)

Take $v = \mathbf{1}$, $w^{\mathsf{T}} = n^{-1}\mathbf{1}^{\mathsf{T}}$, $E = \ker w^{\mathsf{T}}$ and $\Pi_\perp = I - vw^{\mathsf{T}}$. Then $x = r\mathbf{1} + y$ is a unique decomposition with $y \in E$. Every circulant row sum and column sum is $d+C+G-H$, so $J_f E \subset E$. The complexification of E is spanned by the nonuniform Fourier vectors. Its spectrum is strictly left, so its real restriction $B_f = J_f|_E$ is invertible: any real kernel vector would also lie in the zero complex kernel, and injectivity in finite dimension implies surjectivity.

For $T(r, y, g) = \Pi_\perp F(r\mathbf{1}+y, g)$, $D_y T(r_f, 0, g_f) = \Pi_\perp J_f|_E = B_f$. The real IFT gives open neighborhoods I_r, I_g, U_E and one smooth $\psi : I_r \times I_g \to U_E$, with $T = 0$ in their product if and only if $y = \psi(r, g)$. Choose U_E containing zero and shrink the parameter product if necessary. Synchrony supplies $T(r, 0, g) = 0$ for every pair in that same product. Uniqueness therefore forces $\psi(r, g) = 0$ for every pair, not just at the fold.

Every nearby full equilibrium solves $T = 0$, hence has $y = 0$. Its remaining mean equation is $f(r, g) = 0$; for the selected $r > 0$, this is exactly $\nu - Au^2 = 0$. Conversely its roots do solve all components. A state ball around $r_f\mathbf{1}$ times a small g interval fits inside the product. This proves local completeness and excludes hidden nonsynchronous equilibria.

3.2 Nullvectors and original-field derivatives (checks 12–14)

Each shift has both row and column sum one. Therefore

$$J_f v = 0, \qquad w^{\mathsf{T}} J_f = 0, \qquad w^{\mathsf{T}} v = 1.$$

Holding a, k, c, h fixed in (1), $(F_g)_i = -x_i/(1 + x_{i+q})$, hence

$$\alpha_{\mathrm{SN}} = w^{\mathsf{T}} F_g = -r_f/s_f < 0.$$

For $g = g_f - \nu$, $F_\nu = -F_g$ and the coefficient is $+r_f/s_f$.

Along $x = (r_f + t)\mathbf{1}$ the original terms have second derivatives

$$0, \quad -2k, \quad -2c/s_f^3, \quad +2g_f/s_f^3, \quad -2h.$$

In particular $d^2(r/(1+r))/dr^2 = -2/(1+r)^3$ establishes both rational terms independently. Their sum is

$$-2(k+h) + \frac{2(g_f - c)}{s_f^3} = -2A + \frac{2A}{s_f} = -\frac{2Ar_f}{s_f}.$$

It is the same in each of the n components. The mean w^T therefore introduces no additional n factor. The full contraction and half-Taylor coefficient are respectively

$$w^\mathsf{T} F_{xx}[v,v] = -\frac{2Ar_f}{s_f}, \qquad \beta_{\mathrm{SN}} = \frac{1}{2} w^\mathsf{T} F_{xx}[v,v] = -\frac{Ar_f}{s_f} < 0.$$

These computations retain all original terms at every offset collision.

3.3 Extended center dynamics and time orientation (checks 15–18)

In (u, y, ν) coordinates for $(\dot{x}, \dot{\nu}) = (F(x, g_f - \nu), 0)$, $F_\nu = (r_f/s_f)\mathbf{1}$ at the fold, its transverse projection vanishes, and $w^\mathsf{T} J_f = 0$. The linearization is consequently

$$L = \begin{pmatrix} 0 & 0 & r_f/s_f \\ 0 & B_f & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

On the (u, ν) plane it is a nonzero nilpotent block with a double algebraic zero and one-dimensional kernel. This is the *extended* zero multiplicity; the fixed-parameter Jacobian still has a simple zero. The generalized center space is exactly $y = 0$. Unlike P09 at the origin, the parameter derivative here does not vanish.

The set $y = 0$ with variable (u, ν) is smooth and invariant by synchrony and $\dot{\nu} = 0$, and is tangent to the whole generalized center space. It is thus an exact local two-dimensional center manifold. At fixed $\nu = 0$ its state slice is the exact one-dimensional center manifold. This uses invariance and tangency, not equilibrium reduction alone. Its restriction is (3) together with $\dot{\nu} = 0$.

For $|u| < r_f/2$,

$$\frac{r_f + u}{s_f + u} \geq \frac{r_f/2}{1 + 3r_f/2} > 0.$$

Thus $d\tau/dt = (r_f + u)/(s_f + u)$ is smooth and orientation preserving, and division yields exactly

$$\frac{du}{d\tau} = \nu - Au^2, \qquad \frac{d\nu}{d\tau} = 0.$$

Simple fixed-parameter zero, stable complement, nonzero α_{SN} and β_{SN}, local equilibrium completeness, and this exact center dynamics jointly justify the nondegenerate full-system saddle-node label. The side $g > g_f$ has no equilibria in the fold neighborhood; $g = g_f$ has a repeated reduced root; $g < g_f$ sufficiently close has precisely two positive equilibria. "Double equilibrium" here does not assert a double eigenvalue of the fixed-parameter Jacobian. The normal form does not decouple the entire nonlinear system globally.

3.4 The fold itself is not a sink (check 19)

At $\nu = 0$ write $q(u) = (r_f + u)/(s_f + u) > 0$. Then $\dot{u} = -Aq(u)u^2$. Small $u > 0$ decreases to zero, by monotonicity and uniqueness; small $u < 0$ decreases away from zero. On a fixed compact negative interval, $q \geq q_* > 0$, so $v = -u$ satisfies $\dot{v} \geq Aq_* v^2$ until exit. Arbitrarily small negative initial perturbations exit a prescribed small neighborhood in finite time. They lie within the positive orthant for a sufficiently small neighborhood. Thus the fold is not Lyapunov stable, and in particular is not locally asymptotically stable in the open state space. **FULL-SYSTEM SADDLE-NODE VERDICT: VERIFIED.**

4 Layer C: branch stability and open coexistence

4.1 Whole-interval uniform signs (checks 20–21)

For $g_c < g < g_f$,

$$\Delta^2 = b^2 + 4A(a + c - g) = 4A(g_f - g), \qquad 0 < \Delta < b, \qquad r_\pm = \frac{b \pm \Delta}{2A} > 0.$$

Therefore $b - 2Ar_\pm = \mp\Delta$. At the roots the polynomial factor in f vanishes, so differentiation leaves only $f_r = r(b - 2Ar)/(1 + r)$. Hence

$$\lambda_0(r_-) = \frac{r_-\Delta}{1 + r_-} > 0, \qquad \lambda_0(r_+) = -\frac{r_+\Delta}{1 + r_+} < 0.$$

These signs are valid on the entire open interval, not merely asymptotically near the fold. The lower equilibrium is nonlinearly unstable even if another mode happens to be nonhyperbolic; a strictly positive eigenvalue suffices. A loss of upper stability within this interval cannot be a uniform-mode loss.

4.2 Finite-label persistence and dimensions (checks 22–24)

Every labeled $\lambda_m(r, g)$ is smooth near (r_f, g_f), and there are only finitely many labels for fixed n. Intersect their neighborhoods so $|\operatorname{Re}\lambda_m(r, g) - \operatorname{Re}\lambda_m^f| < \delta_f/2$ for $m \neq 0$. Both $r_\pm(\nu)$ are continuous at $\nu = 0$; differentiability in ν is not required. Some $\varepsilon_{\mathrm{sp}} > 0$ therefore puts both branches in this common neighborhood for $0 \leq \nu < \varepsilon_{\mathrm{sp}}$, with

$$\operatorname{Re}\lambda_m(r_\pm, g_f - \nu) < -\delta_f/2 \quad (m \neq 0).$$

Shrink the interval to preserve root positivity. For $\nu > 0$ there is exactly one positive eigenvalue on the lower branch, the uniform one. It is simple because every other label is strictly stable. There are $n - 1$ stable real dimensions and one unstable real dimension. Each nonreal conjugate pair contributes two real dimensions, as do two stable labels with coinciding real eigenvalue; multiplicity is not discarded. The upper branch is Hurwitz, hence locally exponentially stable. No maximizing-mode theorem is involved.

4.3 Origin, parameter openness and meaning of bistability (checks 25–28)

Directly at the origin $J(0) = (a - g)I + cP$, so

$$\operatorname{Re}\lambda_m(0) = a + c - g - c(1 - \cos\theta_m) \leq g_c - g < 0.$$

This is independent of q and proves exponential stability throughout the whole two-positive-root interval, in the local open extension and relative to the invariant nonnegative orthant. Choose any $0 < \varepsilon <$

$\min\{\varepsilon_{\mathrm{sp}}, b^2/(4A)\}$. The interval $(g_f - \varepsilon, g_f)$ is nonempty, lies above g_c, and has the origin sink, lower index-one saddle and upper sink simultaneously.

At a strict witness each Jacobian is invertible and hyperbolic, and the three equilibria are distinct. The IFT continues them smoothly under changes in (a, k, c, g, h). The origin remains exactly zero. The two positive roots are also explicit smooth functions because their discriminant is strictly positive, so local uniqueness identifies these with the IFT continuations. After shrinking a parameter neighborhood, root separation, root positivity, coefficient positivity, $a > k + h$, $g_c < g < g_f$ and all strict spectral signs persist. This is an open set in the five-dimensional physical parameter space for each fixed (n, q), not merely an open segment in g. Dependence of its size on (n, q) is expressly allowed.

P10 defines bistability as coexistence of at least two locally stable equilibria. Its statements are consistent with that definition: they neither count all attractors nor exclude periodic or other invariant attracting sets. **BRANCH STABILITY / BISTABILITY VERDICT: VERIFIED.**

5 Layer D: exact coexistence set and counterexample

5.1 Openness and the two directions of exactness (checks 29–30)

Let

$$\mathcal{D} = \{(a, k, c, g, h) \in (0, \infty)^5 : a > k + h, \quad g_c < g < g_f\}.$$

It is open since the functions and inequalities are continuous and strict. On it $r_+ = (b + \sqrt{b^2 + 4A(a + c - g)})/(2A)$ is smooth: $A > 0$, the radicand is positive, and $1 + r_+ > 0$. Thus each labeled eigenvalue is smooth. The maximum of finitely many continuous real parts is continuous even at mode ties, so

$$\mathcal{B}_{n,q} = \{\pi \in \mathcal{D} : \max_m \operatorname{Re} \lambda_m(r_+, g) < 0\}$$

is open; the near-fold witnesses make it nonempty.

Every member has two local exponential sinks since the origin is Hurwitz throughout $\mathcal{D}$ and so is the upper root on $\mathcal{B}_{n,q}$. For the converse, use the standard meaning of local exponential stability: there exist $K, \eta, \delta > 0$ such that

$$\|\Phi_t(p + v) - p\| \le K e^{-\eta t} \|v\| \quad (t \ge 0, \ \|v\| < \delta).$$

For each fixed t, smooth flow dependence gives $D\Phi_t(p) = e^{tJ}$. Substitute $v = \epsilon z$, divide by $|\epsilon|$, and let $\epsilon \to 0$ to obtain $\|e^{tJ} z\| \le K e^{-\eta t} \|z\|$ with the same constants for all $t \ge 0$. The induced semigroup bound implies $\max \operatorname{Re} \operatorname{spec}(J) \le -\eta < 0$, for example by spectral mapping and the spectral-radius bound on the matrix norm. The upper root is interior to the orthant, so a relative exponential estimate also applies to an ordinary full ball there. This proves both directions of "exactly." Nonhyperbolic asymptotically stable boundary points are not classified.

5.2 The sufficient inequality and its explicit interval (checks 31–32)

At any synchronous state subtract the spectral row sum:

$$\operatorname{Re} \lambda_m = \lambda_0 + C(\cos \theta_m - 1) + (G - H)(\cos(q\theta_m) - 1).$$

Here λ_0 means the uniform row sum; its simplified polynomial-root formula is used only at equilibria. On the upper root $\lambda_0 < 0$. If $G \ge H$, the two remaining contributions are nonpositive. For $m \ne 0$ the C term is strictly negative, and for $m = 0$ the already negative uniform value suffices. Equality $G = H$ is therefore included. Since $r_+ > 0$, $G \ge H$ is equivalent to $g/(1 + r_+)^2 \ge h$.

Put

$$M = g_f - hs_f^2 = c + ks_f^2 > 0, \qquad t_* = \min\left\{\frac{r_f}{2}, \frac{M}{4hs_f}, \sqrt{\frac{M}{2(A+h)}}\right\} > 0.$$

For $0 < \nu < At_*^2$ set $t = \sqrt{\nu/A}$, so $0 < t < t_*$, $g = g_f - At^2$ and $1 + r_+ = s_f + t$. Direct expansion gives

$$g - h(1+r_+)^2 = M - 2hs_f t - (A+h)t^2.$$

The strict inequality $t < t_* \le M/(4hs_f)$ gives $2hs_f t < M/2$; likewise $t < t_* \le \sqrt{M/[2(A+h)]}$ gives $(A+h)t^2 < M/2$. Their sum is strictly less than M, so the displayed quantity is positive. Moreover,

$$At_*^2 \le \frac{Ar_f^2}{4} = \frac{b^2}{16A} < \frac{b^2}{4A} = g_f - g_c.$$

Thus $(g_f - At_*^2, g_f)$ is a nonempty sufficient stable-coexistence interval independent of (n, q). This assertion concerns the two sinks only; to retain lower index one, intersect it with the previous transverse-persistence interval. The source makes that distinction explicitly.

5.3 Exact six-node witness and all real parts (checks 33–36)

For $(n, q) = (6, 1)$ and $(a, k, c, g, h) = (6, 1/10, 1/10, 101/10, 9/10)$,

$$A = 1, \quad b = 5, \quad g_c = 61/10, \quad g_f = 247/20, \quad g - g_c = 4 > 0, \quad g_f - g = 9/4 > 0.$$

The nonzero polynomial is $r^2 - 5r + 4 = (r-1)(r-4)$. Direct substitution in all original summands gives $f(1, g) = f(4, g) = 0$. At $r_+ = 4$,

$$(d, C, G, H) = \left(-\frac{109}{250}, \frac{1}{50}, \frac{202}{125}, \frac{18}{5}\right), \qquad G - H = -\frac{248}{125}.$$

Since $q = 1$, $\operatorname{Re}\lambda_m = d + (C + G - H)\cos(2\pi m/6)$, and mode 3 is real. In label order $m = 0, \ldots, 5$ the complete real-part lists are

$$r_+ = 4: \quad \left(-\frac{12}{5}, -\frac{709}{500}, \frac{273}{500}, \frac{191}{125}, \frac{273}{500}, -\frac{709}{500}\right),$$
$$r_- = 1: \quad \left(\frac{3}{2}, \frac{53}{80}, -\frac{81}{80}, -\frac{37}{20}, -\frac{81}{80}, \frac{53}{80}\right).$$

In particular $\lambda_3(r_+) = d - C - G + H = 191/125 > 0$ whereas $\lambda_0(r_+) = -12/5 < 0$: instability is genuinely nonuniform. For the lower root the independent coefficients are $(d, C, G, H) = (-7/40, 1/20, 101/40, 9/10)$. Its unstable labels are $0, 1, 5$; labels $1, 5$ form a real two-dimensional unstable pair, so total unstable real dimension is three. The remaining three labels are stable.

This refutes universal upper stability and universal lower index one on the whole open interval. It is fully compatible with the near-fold statements. It does not establish a crossing type, nonzero frequency at a crossing, Hopf bifurcation, periodic orbit, or any other attractor. **COEXISTENCE SET / COUNTEREXAMPLE VERDICT: VERIFIED.**

6 Layer E: basin thresholds and the local stable disk

6.1 Global threshold within the synchronous ray (checks 37–38)

For every $g_c < g < g_f$ the exact factorization is

$$f(r, g) = -\frac{Ar}{1+r}(r - r_-)(r - r_+).$$

For positive r the prefactor is negative. Therefore the signs on $(0, r_-)$, (r_-, r_+) and (r_+, ∞) are respectively negative, positive and negative. Smooth uniqueness prevents crossing an equilibrium. P01 supplies global nonnegative solutions and boundedness; here monotonicity also gives direct bounds. A solution in $(0, r_-)$ decreases and is bounded below by zero; one between the roots increases and stays below r_+; one above r_+ decreases and stays above it. Each has a limit. If f at that limit were nonzero, continuity would bound its magnitude away from zero with fixed sign near the limit, contradicting convergence. The initial strict inequalities exclude the wrong endpoint. Hence

initial amplitude	limit
0	0 (constant)
$0 < r(0) < r_-$	0
$r(0) = r_-$	r_- (constant)
$r_- < r(0) < r_+$	r_+
$r(0) = r_+$	r_+ (constant)
$r(0) > r_+$	r_+

These are exact full-system solutions by invariance. The conclusion holds even where the upper root is transversely unstable and makes no statement about nonsynchronous points there.

6.2 Manifold existence and exact unstable germ (checks 39–41)

Fix a near-fold tuple for which $p = r_-\mathbf{1}$ is a hyperbolic index-one saddle and both 0 and $r_+\mathbf{1}$ are sinks. The smooth stable/unstable manifold theorem applies in an open neighborhood of p inside the positive orthant. Its stable tangent space is E with dimension $n-1$ and its unstable space is $\operatorname{span}\{\mathbf{1}\}$ with dimension one.

To establish equality of the unstable germ with synchrony, write $s = r - r_-$ on the synchronous line. There $\dot{s} = s(\lambda + \eta(s))$, with $\lambda = f_r(r_-, g) > 0$ and $\eta(0) = 0$. Choose a small interval $|s| < \rho_0$ contained in $0 < r < r_+$ and with $|\eta| < \lambda/2$. In reverse time $\tau = -t$, $|s|$ decreases at least exponentially at rate $\lambda/2$; it stays in the interval for all $\tau \geq 0$ and tends to zero. The lifted vector trajectory has the same property. Thus every point of this synchronous interval belongs to the local unstable set. The uniqueness and backward-orbit characterization of the unstable manifold, or the local inclusion of two one-dimensional embedded manifolds through p, makes their germs equal. Tangency alone is not the argument.

The left branch stays in $0 < r < r_-$ and has forward limit zero; the right branch stays in $r_- < r < r_+$ and has forward limit r_+. Their negative-time limit is p. They are therefore synchronous heteroclinic trajectories of the original full system. Their forward continuation need not remain inside the chosen local saddle chart.

6.3 A legitimate chart and exact equations (checks 42–45)

By the graph form of the stable manifold, in coordinates $x = p + s\mathbf{1} + y$, $y \in E$, its germ is $s = \phi(y)$, where $\phi(0) = 0$ and $D\phi(0) = 0$. Define

$$\xi = s - \phi(y), \quad z = y, \qquad x = p + (\xi + \phi(z))\mathbf{1} + z.$$

The triangular transformation has an explicit smooth inverse on its chart. The stable disk becomes $\xi = 0$. Since $z = 0$ means $y = 0$ and $\phi(0) = 0$, the synchronous unstable segment remains $z = 0$. At the origin the linearized coordinate transformation is the identity in (s, y), so the linear blocks remain $\lambda > 0$ and $B = J(p)|_E$.

Let $X(\xi, z)$ denote the transformed $\dot{\xi}$. Local invariance of the stable graph implies $X(0, z) = 0$ throughout a small graph domain, not just at $z = 0$. The fundamental theorem of calculus gives the exact factorization

$$X(\xi, z) = \xi \int_0^1 \partial_\xi X(t\xi, z)\, dt = \xi(\lambda + \alpha(\xi, z)), \qquad \alpha(0, 0) = 0.$$

Let $Z(\xi, z)$ denote $\dot{z}$. Invariance of synchrony yields $Z(\xi, 0) = 0$. Therefore with $R = Z - Bz$,

$$\dot{z} = Bz + R(\xi, z), \qquad R(\xi, 0) = 0, \quad D_z R(0, 0) = 0.$$

On a small product neighborhood with transverse balls, the segment from 0 to z remains in the domain and

$$R(\xi, z) = \int_0^1 D_z R(\xi, tz) z\, dt.$$

Continuity of $D_z R$ thus makes $\|R\| \le (\gamma/2)\|z\|$ uniform over all sufficiently small ξ, once $\gamma > 0$ is chosen below. These are exact local equations, not an unproved linear decoupling.

6.4 Why the nonsymmetric transverse matrix contracts (check 46)

The key point is the *orthonormal* Fourier diagonalization. Let Q_E contain the nonuniform Fourier columns. Then on the complexification of E,

$$B = Q_E \operatorname{diag}(\lambda_m(p) : m \neq 0) Q_E^*, \qquad \frac{B + B^*}{2} = Q_E \operatorname{diag}(\operatorname{Re} \lambda_m(p) : m \neq 0) Q_E^*.$$

For real $z \in E$, the antisymmetric part has zero quadratic form. With $\gamma = -\max_{m \neq 0} \operatorname{Re} \lambda_m(p) > 0$ this proves

$$\langle z, Bz \rangle = \sum_{m \neq 0} \operatorname{Re} \lambda_m(p)\, |(Q_E^* z)_m|^2 \le -\gamma \|z\|^2.$$

Although B is generally nonsymmetric, normality and the invariant orthogonal decomposition justify this Euclidean estimate. A general Hurwitz matrix would not justify it. P10 invokes exactly the structure needed.

6.5 Decay, sign and finite exit (checks 47–49)

Shrink the product neighborhood so $|\alpha| < \lambda/2$ and the remainder bound holds. Then along trajectories staying there,

$$\frac{d}{dt}\|z\|^2 = 2\langle z, Bz \rangle + 2\langle z, R \rangle \le -\gamma \|z\|^2, \qquad \|z(t)\| \le e^{-\gamma t/2} \|z(0)\|.$$

For $\xi(0) \neq 0$,

$$\xi(t) = \xi(0) \exp\left(\int_0^t [\lambda + \alpha(\xi(s), z(s))], ds \right).$$

This preserves its sign and gives $|\xi(t)| \ge |\xi(0)| e^{\lambda t/2}$ while in the chart. For $\xi(0) = 0$, uniqueness preserves $\xi = 0$.

Choose $\rho > 0$ smaller than both r_- and $r_+ - r_-$, and small enough for $(\pm\rho, 0)$ and a compact surrounding product to lie within the coordinate domain and the positive orthant. These two points are on the synchronous heteroclinic branches and belong to the basins of 0 and $r_+\mathbf{1}$, respectively. Each sink has a local attracting neighborhood. For either exit point, convergence puts its orbit in that neighborhood at some finite time. Continuous dependence up to that time puts an open neighborhood of the exit point

in the same basin. P01 gives subsequent global nonnegative existence. This proves relative openness of the basins, including for the boundary sink at the origin.

Because the exit points themselves are interior to the orthant and their basins are relatively open, select $\sigma > 0$ sufficiently small that both *closed* caps $\{(\pm\rho, z) : \|z\| \le \sigma\}$ lie in the corresponding basins. The closure of the entire coordinate box can simultaneously be kept inside the chart and positive orthant. The order is to choose the chart, then ρ, then shrink σ for both caps and compact containment.

For $\xi(0) \neq 0$ in $(-\rho, \rho) \times \{\|z\| < \sigma\}$, transverse decay precludes exit through the z boundary. Sign preservation selects a unique ξ cap. If it had not exited by

$$T_* = \frac{2}{\lambda} \log \frac{\rho}{|\xi(0)|},$$

the growth estimate would force $|\xi(T_*)| \ge \rho$, a contradiction. There is no earlier hidden chart exit or finite-time loss of existence: up to a box exit the trajectory remains in a compact subset of the smooth open domain, and ordinary continuation applies. It therefore reaches the signed cap in finite time and subsequently converges to its sink. If $\xi = 0$, it remains there, z decays to zero, and the full state tends to p. The stable disk is forward invariant in this selected box.

6.6 Connected sides and exact relative boundary (checks 50–51)

Let V be the image of the open box and let S be the image of its middle disk $\xi = 0$. The inverse chart proves V is an open neighborhood of p; S is an embedded codimension-one open disk, relatively closed in V. The two half-boxes are nonempty and connected, so their images V_-, V_+ are the two connected components of $V \setminus S$. The preceding exit proof gives

$$V_- \subset \mathcal{A}(0), \qquad V_+ \subset \mathcal{A}(r_+\mathbf{1}), \qquad S \subset \{x : \Phi_t(x) \to p\}.$$

Limits to distinct equilibria are incompatible. Since these three sets exhaust V, in fact $\mathcal{A}(0) \cap V = V_-$ and $\mathcal{A}(r_+\mathbf{1}) \cap V = V_+$. Every point of the middle disk can be approached from either half by varying ξ, while each half is open in V. Hence

$$\partial_V(\mathcal{A}(0) \cap V) = S = \partial_V(\mathcal{A}(r_+\mathbf{1}) \cap V).$$

This proves dynamical two-basin separation, not merely codimension-one topology. It does not identify the complete global boundary, the entire continued stable manifold, or a partition of the whole orthant. Other equilibria and periodic or other attractors elsewhere are not excluded. P10 makes all these exclusions explicitly in lines 619–626. **BASIN / MANIFOLD VERDICT: VERIFIED.**

7 Layer F: P09 connection, degenerate limit and P11 boundary

For fixed $b > 0$, as $g \downarrow g_c$ from above, $\Delta \to b$, so

$$r_- \to 0, \qquad r_+ \to b/A.$$

Since $\Delta'(g) = -2A/\Delta$,

$$\frac{dr_-}{dg} = -\frac{\Delta'(g)}{2A} = \frac{1}{\Delta} \longrightarrow \frac{1}{b}.$$

This is P09's small branch slope in the physical parameter g; its slope in $\mu = g_c - g$ has the opposite sign. For $b > 0$ the small positive branch exists for $g > g_c$ and is exactly the lower root. It remains

uniformly unstable until its collision with the upper root at g_f. No claim gives upper transverse stability all the way to g_c.

Holding k, c, h fixed, let $b = a - A \downarrow 0$. Then

$$r_f \to 0, \qquad g_f - g_c \to 0, \qquad \alpha_{\mathrm{SN}} = -r_f/s_f \to 0, \qquad \beta_{\mathrm{SN}} = -Ar_f/s_f \to 0.$$

At the limiting locus $a = A$, $g = a + c$, the exact center field is $-Ar^3/(1+r)$, with third derivative $-6A$ at zero. Its fixed-parameter center zero is still simple, by the origin gap. The constraints $a - k - h = 0$ and $g - a - c = 0$ have independent gradients in the five physical parameters, consistent with P09's algebraic/spectral codimension-two statement. This limit is excluded from P10's nondegenerate positive-fold theorem. It does not establish a cusp or any two-parameter unfolding.

The P11 input ledger contains only: (i) lower uniform eigenvalue positive throughout the two-root interval; (ii) upper uniform eigenvalue negative throughout; (iii) full upper stability near the fold; (iv) any later upper instability inside the interval must be nonuniform; and the complete fold spectrum supporting these facts. It establishes no Hopf existence, critical-pair simplicity, nonzero crossing frequency, crossing transversality, first Lyapunov coefficient, or periodic branch. P11's own frozen direct inputs remain P02/P04/P05; P10's future-input language is not a graph amendment. After administrative A10 closure, the frozen attempt order permits P11 to begin its own scoped work. This audit neither begins it nor verifies any of its proposed theorems. **SCOPE / DEPENDENCY VERDICT: VERIFIED.**

8 Active adversarial tests and independent corroboration

8.1 Attempts to break the mathematical claims (checks 55–56)

The following are analytical attacks on specific vulnerable steps, not inferences from a failed numerical search. Their resolutions refer to the independent arguments above.

No.	Attack	Resolution within the stated hypotheses
1	Try to make $G_f - H_f \le 0$ by increasing h.	The physical fold relation forces $g_f/s_f^2 - h = k + c/s_f^2 > 0$; h is not freely varied while holding the fold's g, r fixed.
2	Kill the transverse cosine loss with noncoprime q.	Even when $\cos(q\theta_m) = 1$, $C_f(\cos\theta_m - 1)$ is strictly negative. Removing $c > 0$ would leave the canonical regime.
3	Create another zero via mode collisions or parity.	Unitary diagonalization plus the strict margin leaves only the uniform label at zero; stable repetitions are harmless.
4	Hide a nonsynchronous nearby equilibrium.	The real transverse IFT has a unique solution in a product containing the known zero transverse solution for every nearby (r, g).
5	Find an n or $1/2$ coefficient error.	All five original second directional derivatives are equal componentwise; averaging cancels n, and the Taylor half is applied exactly once.
6	Reverse time through a vanishing prefactor.	The neighborhood $\lvert u\rvert < r_f/2$ gives a strictly positive uniform lower bound. The $b = 0$ limit is explicitly excluded.
7	Extend index one too far from the fold.	It is asserted only in the finite-label persistence neighborhood. The exact interior example has lower index three.
8	Destabilize the upper branch arbitrarily near a fixed fold.	Continuity and the finite transverse margin prevent this; the uniform eigenvalue is strictly negative on the upper root. No uniform-in-n gap is claimed.
9	Destroy openness through a branch collision.	The witness is strictly inside the interval, all equilibria are distinct and hyperbolic, and the IFT neighborhood excludes both endpoints.

No.	Attack	Resolution within the stated hypotheses
10	Break sufficiency at $G = H$.	The q term vanishes, leaving negative λ_0 and nonpositive step-one correction. Equality is allowed without a limit argument.
11	Make the explicit interval empty or lose strictness.	Every defining bound is positive. The open interval gives strict inequalities for both $M/2$ estimates, even if the minimum has ties.
12	Invalidate the exact counterexample arithmetic.	Rational calculation gives $\lambda_3(r_+) = 191/125$ and lower real parts $3/2, 53/80, -81/80, -37/20, -81/80, 53/80$.
13	Cross a scalar threshold or converge to a nonzero velocity.	Uniqueness prevents equilibrium crossing; monotone bounded orbits have a limit that must be a scalar zero.
14	Replace an unstable manifold by a merely tangent curve.	Every nearby synchronous point has a full backward exponentially convergent orbit. This proves membership and equality of germs.
15	Escape sideways, violate Euclidean contraction, or land in a third basin.	The Fourier symmetric-part identity gives contraction, the invariant-coordinate factorization gives signed finite cap exit, and the two closed caps lie in sink basins. The entire local box is thereby classified.

The attacks identify where each strict hypothesis matters. They produce no counterexample to the statements actually made. The failure of broader whole-interval spectral conclusions is independently exhibited, not hidden. The local manifold and basin proofs are analytic; no numerical approximation to a stable disk is offered as a certificate.

8.2 Independent exact and direct-matrix checks

The new `A10_support/independent_checks.py` was written from the original five-term field, without importing, executing or using the developer checker to generate it. Its final run uses Python 3.12.14 and NumPy 2.3.5 and passes:

- 794 exact rational assertions covering six positive-b folds, square completion, original scalar residuals, both branch signs, time-factor bounds, fold coefficients, explicit-interval inequalities, ray signs, and the exact six-node counterexample. An additional exact $G = H$ upper-root witness uses $(a, k, c, g, h) = (23/5, 1/10, 3/10, 81/10, 9/10)$ with $r_+ = 2$, $r_- = 8/5$ and $\lambda_0 = -4/15$.
- 15,651 original-field/direct-matrix cases across all 139 admissible (n, q) pairs for

  $$n \in \{5, 6, 7, 8, 9, 10, 12, 15, 16, 18, 21, 24\}.$$

  This includes $q = 1$, $q = n - 1$, even $q = n/2$, noncoprime cases, 1,668 near-fold branch cases, five deeper branch distances per fixed tuple, explicit sufficient-interval checks and $G = H$ cases. The near-fold distance is shrunk when needed for the fixed transverse gap; this is finite corroboration of a neighborhood, not a computed universal radius.
- 1,500 additional static random interior parameter tuples, included in the matrix total with both branches. The fixed seed is 100310. There are 169 sampled unstable-upper points and 536 sampled lower points with more than one unstable label. The conditional upper criterion $G \geq H$ fails in none of its sampled applicable cases. These counts are search outcomes, not classifications of parameter regions.
- All direct matrix entries are assembled by separately differentiating and adding the original coordinate terms. A perfect bipartite matching compares all n direct eigenvalues with all n predicted labels,

preserving multiplicity. The symmetric-part spectrum is checked separately. For every (n, q) pair, the direct Jacobian at a nonsynchronous state is also compared with complex-step differentiation of the original field.

The maximum scaled discrepancies are 4.915×10^{-14} for the full spectrum, 2.693×10^{-16} for equilibrium residuals, 1.648×10^{-14} for the symmetric-part spectrum, and 4.086×10^{-16} for the independent Jacobian differentiation check. Exact results and full case records are preserved in JSON/CSV. Floating-point comparisons use explicit tolerances; they are not interval arithmetic and do not prove equilibrium completeness, manifold existence or any basin theorem.

8.3 Validation provenance and limitations

The original `P10_validation_support.zip` was located and its CRC and member hashes recorded. Its archive SHA-256 is

`3b545b222be77d2e8714784da34cea6494b4739b5241c3c60cd697ac69f4d760`

It remains unchanged. Its checker was not rerun, and this audit does not claim to reproduce its historical run totals or outputs. Its contents were inventoried only after the independent checker was written and run. The manifest's developer-validation statistics remain developer records; none enters the mathematical verdict. No original developer source or output is copied into this audit-only delivery as independent evidence.

During construction of the new checker, an optional SciPy import was unavailable; it was replaced by a local multiplicity-preserving matching routine using NumPy. A mistyped *new audit-only* equality witness was caught by an exact scalar-residual assertion and corrected to the tuple above before the final complete run. These were checker-construction issues, not defects in P10. The supplied exact P10 witness passed. The final delivered checker and results correspond to the successful complete run; abandoned partial runs supply no evidence. Compilation and PDF inspection below are presentation checks only.

9 Complete line-by-line disposition

The following 62 contiguous ranges cover every line 1–707 exactly once, including structural and blank lines. Each substantive range has exactly one of the requested dispositions: PASS, MINOR ISSUE, MAJOR ISSUE, FALSE, or UNRESOLVED. All ranges are PASS. Detailed arguments appear in Sections 2–7; the CSV copy is `A10_support/line_by_line_review.csv`. No silent correction to the audited text is incorporated into these verdicts.

Lines	Verdict	Finding
1–18	**PASS**	Header correctly fixes DRAFT v1.0, physical parameter g, direct and inherited inputs, S4 role, and local basin exclusions. NEXT DEPENDENCY is a future-work pointer, not a changed DAG edge.
19–29	**PASS**	Canonical positive coefficients, n>=5 and all 1<=q<n; b>0 is explicit. Fold data are physically admissible.
30–36	**PASS**	P02 amplitudes s_-/s_+ become r_-/r_+; its fold amplitude s_f becomes r_f. P10 s_f is the denominator. Dimension-dependent neighborhoods are disclosed.
37–44	**PASS**	Only P02/P03/P04/P09 plus inherited P01 and standard local ODE tools are used. No P05/P07/P08 or future theorem is imported.
45–55	**PASS**	Substitution in all five original summands gives the exact invariant scalar field on r>-1.

Lines	Verdict	Finding
56–63	**PASS**	Nonzero equilibrium polynomial and both fold identities follow by expansion; the gap is strictly positive for b>0.
64–75	**PASS**	With u=r-r_f and nu=g_f-g the mixed linear term cancels exactly, leaving nu-Au^2. No Taylor truncation occurs.
76–87	**PASS**	Positive local prefactor; zero, repeated, or two real roots on the stated parameter sides. The two distinct roots are positive exactly for 0<nu<b^2/(4A); the origin is outside the fold neighborhood.
88–102	**PASS**	Original derivative yields d,C,G,H and the stated predecessor/successor phases. Coincident offsets add, preserving all contributions.
103–110	**PASS**	The simplified uniform formula is used at positive equilibria only. Differentiating the exact scalar field gives zero at the fold.
111–125	**PASS**	Structural identity and complete complex spectrum are correct. Imaginary H contribution has positive sine sign.
126–136	**PASS**	Strict uniform-in-label transverse bound holds for each fixed n. Only mode zero lies on the imaginary axis, with algebraic and geometric multiplicity one.
137–145	**PASS**	Physical relation g_f=c+A*s_f^2 gives G_f-H_f=r_f*(k+c/s_f^2)>0 directly.
146–160	**PASS**	Subtracting the uniform row sum and using j=min(m,n-m) proves the bound for every parity, collision and noncoprime case. Repeated stable values do not affect the simple zero.
161–174	**PASS**	The real mean-zero subspace is invariant and has precisely the nonuniform spectrum on complexification. Invertibility over the reals follows.
175–188	**PASS**	The transverse IFT is applied on a product neighborhood containing y=0 for all nearby (r,g). Invariance and uniqueness force psi identically zero. A smaller state ball and parameter interval give full local completeness.
189–207	**PASS**	Both normalized nullvectors, nonzero parameter derivative, full second contraction, half-Taylor coefficient and reversed nu sign are correct.
208–213	**PASS**	Circulant row and column sums both vanish. Averaging F_g produces -r_f/s_f with no missing column condition.
214–226	**PASS**	Five original directional second derivatives sum to -2A*r_f/s_f. Averaging removes n and Taylor expansion alone supplies the factor one-half.
227–245	**PASS**	Full-system theorem follows with simple zero, stable complement, nonzero coefficients and local equilibrium completeness. Double means a repeated reduced root, not a double zero eigenvalue.
246–252	**PASS**	Smoothness, positivity and all nondegeneracy hypotheses hold at the fixed fold; no nonsynchronous local branch is assumed away.
253–268	**PASS**	Extended linearization has a nontrivial nilpotent two-dimensional center block and stable E block. Invariant synchrony times the parameter interval is tangent to the generalized center subspace.
269–276	**PASS**	Displayed lower bound makes the time change positive. It gives the exact center normal form and asserts no global full-space decoupling.
277–280	**PASS**	At nu=0 the center field is negative on both sides: attraction from above, escape from below. This rules out full-space local asymptotic stability.
281–300	**PASS**	Discriminant and uniform branch signs hold on the entire open two-positive-root interval. Upper instability there must involve a nonuniform mode.
301–309	**PASS**	Exact substitution proves the signs. A positive eigenvalue proves lower instability even if other eigenvalues are nonhyperbolic.
310–323	**PASS**	Near-fold transverse persistence gives the upper sink and lower hyperbolic index-one saddle with n-1 real stable dimensions.
324–338	**PASS**	Finite-label continuity is sufficient despite square-root parameter dependence. Numerical repetition among stable eigenvalues does not change dimensions or uniform eigenvalue simplicity.
339–359	**PASS**	Coexistence theorem quantifies a nonempty near-fold interval and an open five-parameter neighborhood. Bistability explicitly means at least two local sinks.

Lines	Verdict	Finding
360–370	**PASS**	Origin spectrum is strictly left for every g_c<g<g_f. Choosing positive epsilon smaller than both bounds yields a genuinely nonempty interval.
371–384	**PASS**	Distinct hyperbolic roots persist by IFT, with physical positivity and strict parameter inequalities preserved. Dependence on fixed (n,q) is stated.
385–412	**PASS**	D and B_nq have the asserted exact exponential-stability meaning; no whole-interval lower index claim. The G>=H criterion includes equality.
413–419	**PASS**	Upper root and labeled eigenvalues are smooth on D; strict finite maximum defines an open nonempty set. Origin stability is automatic there.
420–425	**PASS**	Differentiating a uniform local exponential flow estimate gives an exponential bound for exp(tJ), proving the converse. Nonhyperbolic asymptotic boundary stability is excluded from the claim.
426–439	**PASS**	Subtracting the row sum is valid at any synchronous state; the simplified equilibrium uniform formula is not used off equilibrium. Nonpositive cosine corrections prove sufficiency including G=H.
440–449	**PASS**	M=c+k*s_f^2 is positive and every defining t_* bound is positive. This interval is dimension-independent for the two sinks only.
450–456	**PASS**	With t<t_*, each subtracted term is strictly less than M/2, so g-h*(1+r_+)^2>0.
457–464	**PASS**	A*t_*^2<=b^2/(16A)<g_f-g_c. Intersecting with the separate spectral neighborhood, rather than this bound alone, retains index one.
465–474	**PASS**	Exact six-node tuple is canonical and strictly within the two-root interval; polynomial is (r-1)(r-4).
475–486	**PASS**	Upper residual vanishes; all four rational coefficients are correct, lambda_0=-12/5 and lambda_3=191/125>0.
487–497	**PASS**	At the lower root the real parts are 3/2, 53/80, -81/80, -37/20, -81/80, 53/80. Unstable real dimension is three. Counterexample is static and makes no Hopf or extra-attractor assertion.
498–516	**PASS**	Ray threshold handles both equilibria and zero explicitly and remains true even with upper transverse instability.
517–533	**PASS**	Exact factorization gives all signs. Uniqueness, monotonicity, bounded global existence and exclusion of a nonzero limiting velocity prove every stated ray limit.
534–545	**PASS**	Under the near-fold hypotheses the stable disk has codimension one and the unstable germ is synchronous. Its two continued branches have the stated full-system heteroclinic limits.
546–558	**PASS**	The stronger partition is explicitly confined to V inside the positive orthant, with basins relative to the nonnegative orthant and boundary relative to V.
559–574	**PASS**	Local backward exponential convergence on the repelling scalar interval, not tangency alone, identifies the unstable-manifold germ. Choose the interval inside (0,r_+) to obtain both limits.
575–582	**PASS**	Stable graph exists over E because its tangent is E. The triangular coordinate change is a smooth local diffeomorphism; z=0 remains the synchronous unstable segment.
583–589	**PASS**	Hadamard integral factorization gives exact divisibility of dot-xi by xi. Invariance of z=0 gives R(xi,0)=0; unchanged derivative at the origin gives D_z R(0,0)=0.
590–593	**PASS**	Unitary Fourier diagonalization controls the symmetric part of B on E; Hurwitz alone would not suffice. Continuity of D_z R gives the uniform remainder estimate after shrinking.
594–596	**PASS**	Squared-norm differential inequality gives transverse decay. The scalar multiplicative equation preserves sign and grows absolute xi at least exponentially.
597–604	**PASS**	Choose rho within both synchronous branches and a compact chart inside the positive orthant, then sigma using openness of each sink basin. Both closed caps can be included.
605–610	**PASS**	Transverse decay prevents a z-face exit; xi growth forces a finite-time signed cap exit. Compact containment excludes any other earlier exit. On xi=0, z tends to zero.
611–618	**PASS**	Connected nonempty half-boxes yield the two sides. S contains neither sink basin, every S point borders both halves, and the whole box is accounted for; relative basin boundaries equal S.

Lines	Verdict	Finding
619–627	**PASS**	Explicit global exclusions prevent interpreting this local partition as the full orthant separatrix or as exclusion of other attractors.
628–640	**PASS**	As g decreases to g_c, lower root tends to zero, upper to b/A, and dr_-/dg=1/Delta tends to 1/b. Identification with P09 has the correct orientation.
641–652	**PASS**	At fixed k,c,h, b tends to zero with both saddle-node coefficients vanishing. The limit matches P09 cubic degeneracy and simple center zero; no cusp/unfolding classification is inferred.
653–662	**PASS**	P11 receives branch signs and fold/near-fold spectral facts only. Hopf existence, pair simplicity, transversality, Lyapunov coefficients and periodic branches remain unproved. Frozen dependency DAG is preserved.
663–675	**PASS**	S4 and B03 are restricted to coexistence and the proved local basin statement. Checker is expressly corroborative and P10 remains DRAFT.
676–683	**PASS**	Audit priorities for notation, fold positivity, gap, parity, collisions and multiplicities are correctly identified and addressed by A10.
684–690	**PASS**	Audit priorities for real IFT, normalized original-vector derivatives, nilpotent center block and positive time factor are addressed.
691–698	**PASS**	Audit priorities distinguish whole-interval uniform signs from near-fold full stability and include openness, equality sufficiency and exact counterexample arithmetic.
699–707	**PASS**	Audit priorities demand backward convergence and cap-exit basin membership, while preserving the local and P11 boundaries. All are addressed.

10 Request coverage and authenticated input ledger

Request checks	Audit disposition and location
0–1	Current-file and dependency authentication in Section 1; full inventories accompany this audit.
2	Six separate layer verdicts on the first page and in Sections 2–7.
3–9	Exact fold geometry, physical identity, phases, gap and multiplicities in Section 2.
10–19	Real transverse IFT, original derivatives, nilpotent center block, exact center flow and fold instability in Section 3.
20–28	Whole-interval uniform signs, near-fold classification, origin and five-parameter openness in Section 4.
29–36	Exact coexistence set, exponential-stability converse, sufficient interval and exact witness in Section 5.
37–51	Ray threshold, unstable germ, smooth chart, contraction, cap exit and exact relative boundary in Section 6.
52–54	P09 orientation, degenerate limit and strictly limited P11 ledger in Section 7.
55–56	Independent finite checks and all 15 explicit adversarial attacks in Section 8.
57	Contiguous 707-line disposition in Section 9 and the companion CSV.
58–60	Exact audit slug, seven verdicts, separate source/PDF and audit-only archive; no replacement proof or full audited-project ZIP.
61–62	Explicit final handoff below; deliver source, reading copy and exactly named audit-only ZIP with concise verdict summary.

The following full-file hashes authenticate the ten control files, sixteen prerequisite proof/audit files, P10 and the additional scope-history record. Hashes identify bytes, not proof validity. Body-hash checks for all eight prerequisite proofs and hashes for every original archive member are also preserved in the support directory.

project_manifest.md
153,664 bytes; 830 lines. Full SHA-256:
b77f9636d7d0ae031b0a4a7127bb804429be21050f44b735d5128730408a57ea

claim_registry.md
113,578 bytes; 571 lines. Full SHA-256:
28f5cd179bbd16889ed9404c8665ed03b6ed2da46d114ef4ad5d2f1d28b10ec5

frozen_theorem_package.md
9,729 bytes; 69 lines. Full SHA-256:
724adf3d04a7f9661a2e53fbae359192e8a5a5ee56737bdca2379209b556ea63

final_novelty_verdict.md
6,511 bytes; 37 lines. Full SHA-256:
5fa8ddd5ada9628b104dba68dd535356442c5e62ec45b72f3c2c354dec6037ea

novelty_decisions.md
19,241 bytes; 124 lines. Full SHA-256:
8fd079fb108b93c4504bc2b527de11c1f7e867a967c7cef0d90e1ff7a0a9175e

02_theorem_dependency_map.tex
9,372 bytes; 145 lines. Full SHA-256:
a1a4435f36fc1d1b6f0b2c5226a913283c76658d55b96493e6c50a8693831cac

proofs_compendium.tex
11,104 bytes; 208 lines. Full SHA-256:
7f06955c706b40d7bee6ca88bd08d8aa99e60c4d5b276536fc33579a9f58a282

shared_macros.tex
1,281 bytes; 35 lines. Full SHA-256:
f4a18a440e1214b619bf18f4a81fee507239460ca796d8ca3916791069d6a7ad

00_canonical_model.tex
3,712 bytes; 74 lines. Full SHA-256:
c209981cb2924e8a782f24aa2904e8c85a105ff99319dd6aada12f48a351c2a4

01_notation_and_assumptions.tex
5,152 bytes; 103 lines. Full SHA-256:
97d993c6d3986524dbcc29659ee73d257be1b03a35c44401e8ef4534303eeb91

proofs/P01_wellposedness_and_absorbing_box.tex
22,570 bytes; 520 lines. Full SHA-256:
0d6950821aa83f3d09e8bfe84dab2b0b5020aa6e7881cf6928e984bff1890f1e

audits/A01_wellposedness_and_absorbing_box.tex
48,622 bytes; 974 lines. Full SHA-256:
8d5c1b2ec782dbe49b07d2bbe529ee474d4c142695c53b523ccb41d63bb33098

proofs/P02_synchronous_subspace_and_equilibria.tex
26,905 bytes; 610 lines. Full SHA-256:
8e057c2cd009659eee45906844e25f4f38c156e5eb2e4130e824aa5e8299c51a

audits/A02_synchronous_subspace_and_equilibria.tex
37,895 bytes; 847 lines. Full SHA-256:
ca3d723d9fcfac14f5129583441e611a65e41848ec44f3b5e3a0072f795d84ac

proofs/P03_origin_and_linearization.tex
19,857 bytes; 447 lines. Full SHA-256:
6c3c9bfb2307bdc043eb7f6a874cb222e2ac2ce0750b339181f0088dd0bd859e

audits/A03_origin_and_linearization.tex
37,700 bytes; 692 lines. Full SHA-256:
e1008f98ba9667b92b8d74f1bc7d21e021414b454ac976ae0db1500b4b7ef8d1

proofs/P04_arbitrary_nq_fourier_spectrum.tex
26,980 bytes; 590 lines. Full SHA-256:
1331c8598acd0481046250d918cac028ca6d13d2604612448c3459e864321c58

audits/A04_arbitrary_nq_fourier_spectrum.tex
47,595 bytes; 832 lines. Full SHA-256:
8a57079241269c92a3c3e8dc4dccf8592da2d1a636fca0c0990a4f318396d869

proofs/P05_q2_mode_selection.tex
28,334 bytes; 613 lines. Full SHA-256:
054a047a55ef37815485f6c225be2f6322ca8cf1c272614dd1dc4bb10ae8e1dd

audits/A05_q2_mode_selection.tex
49,526 bytes; 996 lines. Full SHA-256:
0f7e9836b91009a201185a4acbbb99715f3c65df715e405962b7d6852ff27031

proofs/P07_five_node_specialization.tex
25,691 bytes; 580 lines. Full SHA-256:
1258641da2b1b729523da6262cf75345f5ef1eabf7a9f693a8b13dd34deec393

audits/A07_five_node_specialization.tex
38,488 bytes; 724 lines. Full SHA-256:
b3923d6520cc249c508b80dd06a2b515f054e1510e255224a864e986cd38ce4b

proofs/P08_matched_bilinear_analysis.tex
40,836 bytes; 871 lines. Full SHA-256:
81d273f0bad1714d2470e9fc826f0819e9dde367a9c42633705a275b404e5454

audits/A08_matched_bilinear_analysis.tex
57,343 bytes; 1161 lines. Full SHA-256:
87cd4a03061648eb4cf1c65d1afa47c7132c1f7eed10d546784ee166c8e5c935

proofs/P09_transcritical_bifurcation.tex
25,693 bytes; 564 lines. Full SHA-256:
5d357a5be97dfbad62a0107ca2e010ff97e0c11a719ee850df69411275f11ff8

audits/A09_transcritical_bifurcation.tex
51,832 bytes; 975 lines. Full SHA-256:
1234e02d7cf620511bfd7b9306dbb72ac68929ef01e8c3ba70bd3615bcd2b943

proofs/P10_saddle_node_and_bistability.tex
32,560 bytes; 707 lines. Full SHA-256:
a28ea2b4ca96f381f06fa82849d4289fcc0e9c41e646ca66595621922f385a07

open_decisions.md
17,320 bytes; 101 lines. Full SHA-256:
506c634c9958d13b4d6ef4080ebe214af5fd0c77d5dc8fe6f1b91d2750ffab59

11 Closure handoff to the original P10 development chat

OVERALL VERDICT: VERIFIED. Required corrections: none. There are no E01, E02 or other required mathematical or wording issues to close. This verdict applies only to P10 v1.0 with full-file SHA-256

`a28ea2b4ca96f381f06fa82849d4289fcc0e9c41e646ca66595621922f385a07`

The **original P10 development chat may perform administrative closure without changing the mathematical body**. It should preserve every byte from the first line-anchored section command (line 19) through EOF; the corresponding body hash is

`5cfd8afb91b7f23575e1b4bd9c7c28f3b433affefae9bb0d5f7700d030fcba81`

It may integrate this immutable audit and necessary lifecycle bookkeeping under the established project convention, preserving the original draft archive and all frozen dependencies. It should retain the local basin scope, the whole-interval counterexample and the distinction between audit verdict and administrative lifecycle closure. A change to the mathematical body would require review of that changed version; it is unnecessary for this audit.

After successful correction/closure, the original P10 development chat must create the COMPLETE full-project archive

`cyclic_network_project_P10_audited.zip`.

For this VERIFIED result, the correction step is empty and only administrative closure is needed. **That complete archive is not created in this audit chat.** This chat delivers only `cyclic_network_project_A10_audit_only.zip`, with sole root `cyclic_network_project/`, containing the A10 source and PDF plus its identity, input inventories, complete line review, independent checker and results, provenance and presentation/preservation records. It contains no replacement P10 source. The original `cyclic_network_project_P10_draft.zip` is preserved unchanged.

P11 clearance after closure: yes. Once the original chat records successful P10 administrative closure, P11 is cleared to begin under the frozen attempt order and its own prerequisites and statement gates. This is permission in the research sequence, not a Hopf theorem. No P11 proof, continuation study, later module, two-parameter unfolding, literature survey, or global basin classification has been begun here. P10's DRAFT lifecycle records have not been promoted by this audit.